

Ans to the ques. no 4

$$P(10, 20, -40)$$

$$\beta = 30^\circ$$

$$\lambda = 0.5$$

Here,

$$x' = 7$$

$$y' = y + \lambda \cos \beta x$$

$$z' = z + \lambda \sin \beta x$$

$$\omega = 1$$

$$\therefore \begin{bmatrix} x' \omega \\ y' \omega \\ z' \omega \\ \omega \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \cos 30^\circ \\ 0.5 \sin 30^\circ \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 7 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 \\ 24.33 \\ -37.5 \\ 1 \end{bmatrix}$$

$$\therefore P' = (7, 24.33, -37.5)$$

Ans. to the ques. no. 7

Q $P(10, 20, -40)$

$$\lambda = 0$$

Here,

$$x' = x$$

$$y' = y$$

$$z' = 0$$

$$w = 1$$

That is -

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 \\ 20 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore P' = (10, 20, 0)$$

b) Here,

$$\lambda = 0$$

Now,

$$y' = 13$$

$$z' = z$$

$$x' = x$$

$$w' = 1$$

That is -

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 13 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 \\ 13 \\ -40 \\ 1 \end{bmatrix}$$

$$\therefore P' = (10, 13, -40)$$

Ans. to the ques. no 8

8 Cavalier projection is called full length projection because the lines extending along the ~~receding~~ receding axis are rendered at their exact, true length without any foreshortening.